PROJECT 2 REPORT

REAL AND SYNTHETIC FACE CLASSIFICATION

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Abstract

This project developed is a classification model to distinguish real human faces from synthetic faces. A balanced dataset of 200 images was used, with 100 real images and 100 synthetic images. Images were resized to 128 by 128 pixels and represented using two feature methods: Histogram of Oriented Gradients (HOG) and raw colour pixels. K-Nearest Neighbours (KNN) and Support Vector Machine (SVM) models were trained and evaluated. Cross-validation was used to select the best K value and to compare the four model-feature combinations. HOG-SVM achieved the strongest test accuracy at 92.50% and the highest five-fold cross-validation mean at 93.50%. On 10 new real-world images, Raw-KNN performed best with 80.00% accuracy. The results show that HOG features worked well with SVM on the original dataset, while raw colour features generalised better to the small set of new images.On 10 new images, Raw-KNN performed best at 80.00%. The results show that the best feature type depends on the model and that strong test accuracy does not always guarantee the best performance on unfamiliar images.

1. Introduction

Synthetic face images are becoming more realistic, making it difficult to identify whether a portrait is real or computer-generated. The aim of this project was to build a machine learning classifier that separates real faces from synthetic faces. KNN classifies a new image using nearby labelled examples, while SVM searches for a decision boundary between the two classes. Both algorithms were compared to determine which was more accurate and consistent. HOG and raw colour features were also compared. HOG was chosen to capture facial outlines, edges and texture, while raw pixels were included because the two image categories appeared to contain noticeable colour differences. This tested whether shape-based information or preserved colour information was more useful.

2. Dataset Description

The dataset contained 200 JPG images organised into two folders. The RealFace folder contained 100 real face images and the SynFace folder contained 100 synthetic face images. A separate real-world test set contained five new real images and five new synthetic images. These 10 images were not included in model training. This allows the models to be checked on images from outside the original dataset rather than only on a random test split.

3. Image Preprocessing

Each image was loaded and resized to 128 by 128 pixels so that all images had the same dimensions. Pixel values were divided by 255 to normalise them from the original 0 to 255 range into a 0 to 1 range. For HOG extraction, images were converted to grayscale. Raw features kept the three colour channels and were flattened into one long vector containing 49,152 values.

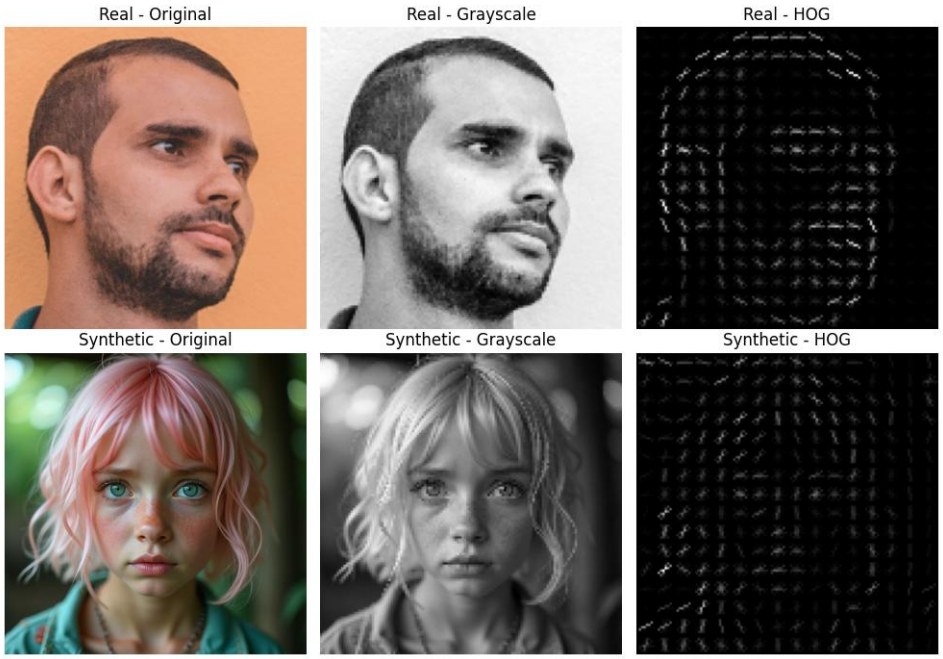


Figure 1. Original, grayscale and HOG representations of one real and one synthetic image.

4. Feature Extraction

Two feature sets were compared to test whether shape or colour was more useful for this dataset. HOG measured edge direction and strength and produced a feature matrix with a shape of (200, 8100). As shown in Figure 1, HOG highlights facial contours, hair boundaries and texture changes while removing colour information. These patterns may reveal unnatural edges or overly smooth regions in synthetic faces. Raw colour pixels produced a larger matrix with a shape of (200, 49152) and preserved the complete resized colour information. I included raw pixels because the real and synthetic images appeared to use contrasting colour tones, lighting and saturation. I expected these differences might make classification easier. Comparing both feature sets therefore tested this idea instead of assuming that edge features would always be better.

5. Model Development

The data was divided into 80% training data and 20% testing data using a fixed random state of 42. For KNN, odd K values from 1 to 49 were evaluated using 10-fold cross-validation on the training data. The K with the highest mean validation accuracy was automatically used to train the final KNN model. The best K was 1 for HOG features with a mean cross-validation accuracy of 80%. The best K was 17 for raw colour pixels with a mean cross-validation accuracy of 82.50%. The SVM model used a linear kernel and balanced class weights.

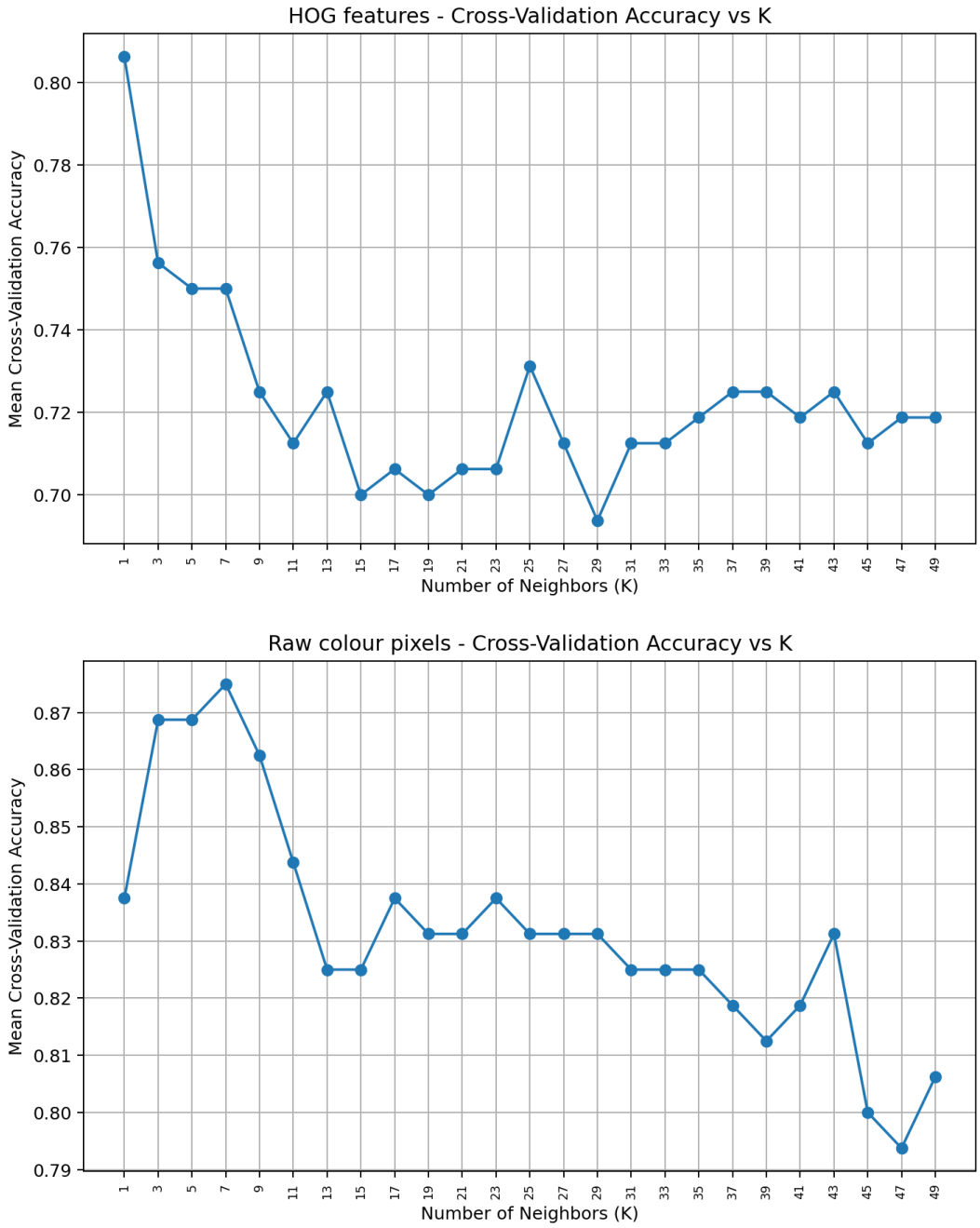


Figure 2. Cross-validation accuracy across K values for HOG and raw colour features.

6. Evaluation

The final models were evaluated using test accuracy and confusion matrices. HOG-KNN achieved 80.00%, HOG-SVM 92.50%, Raw-KNN 82.50%, and Raw-SVM 90.00%. HOG-SVM was therefore the strongest model on the held-out test set. In Figure 3, diagonal values are correct predictions and off-diagonal values are errors. HOG-SVM has fewer off-diagonal errors than HOG-KNN, supporting its higher accuracy. The raw-feature confusion matrices also show that SVM made fewer errors than KNN on the original test split

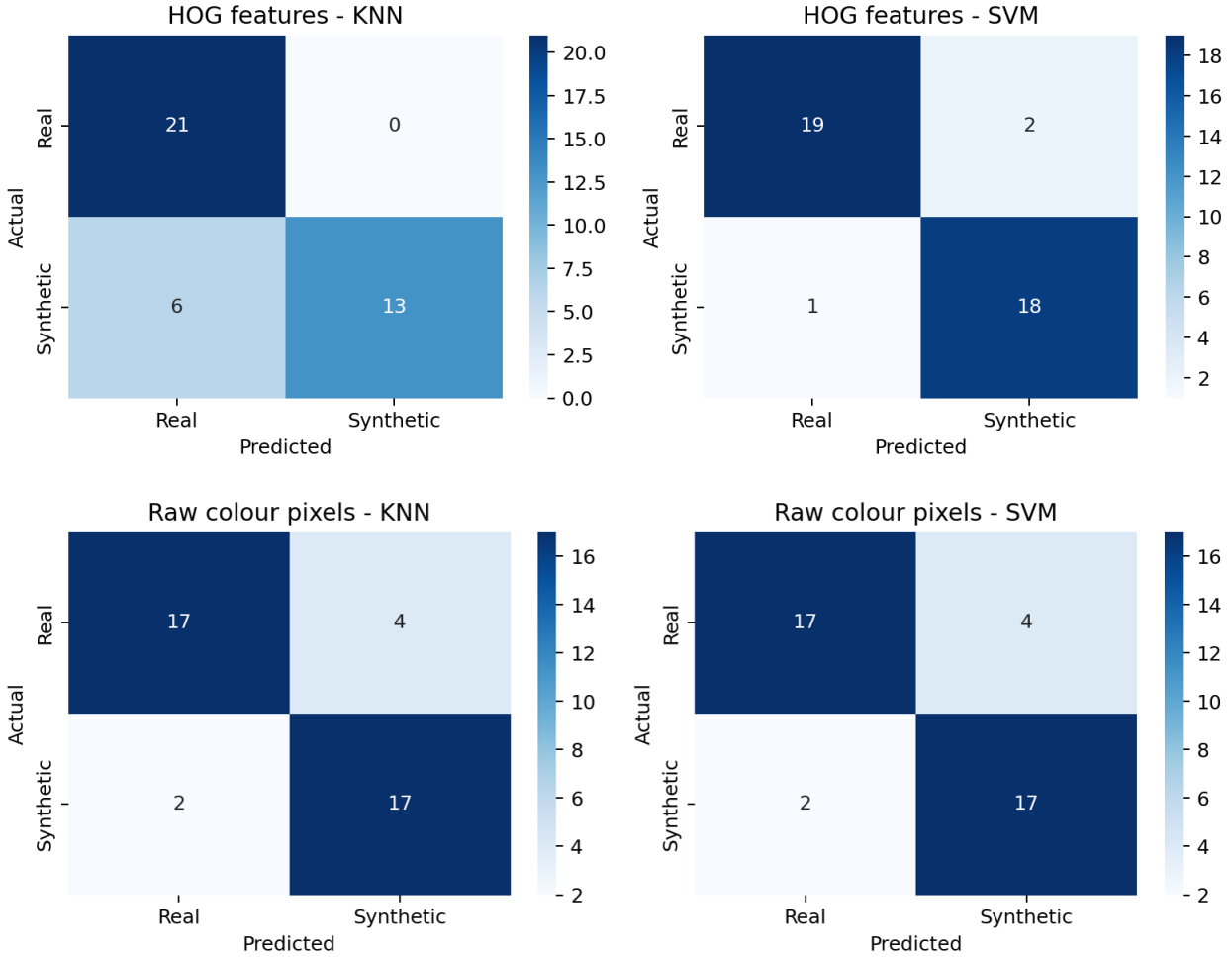


Figure 3. Confusion matrices for KNN and SVM using HOG and raw colour features.

7. Results and Discussion

The five-fold cross-validation comparison in Figure 4 supported the test results. HOG-SVM had the highest mean accuracy at 93.50% with a standard deviation of 3.74%, showing strong performance across different folds. Raw-KNN had the lowest standard deviation at 3.32%, so it was the most consistent, but its lower mean of 84.50% means that consistency alone did not make it the best overall model. Raw-SVM achieved 88.50%, while HOG-KNN achieved 77.50%. SVM likely performed better because it learned a boundary using all training samples, while KNN depended on local similarity and could be affected by one visually similar neighbour. The feature comparison partly supported my colour hypothesis. Raw pixels improved KNN from 80.00% to 82.50% and later achieved the best real-world result. HOG improved SVM from 90.00% to 92.50%, showing that edge and texture information was more useful for SVM on the original dataset.

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HOG -> KNN: 80.00% SVM: 92.50%
Raw -> KNN: 82.50% SVM: 90.00%

Same model, different feature:
KNN: Raw colour better (82.50% vs 80.00%)
SVM: HOG better (92.50% vs 90.00%)
HOG-KNN: [0.725 0.8   0.8   0.725 0.825]
Mean = 77.50% Std = 4.18%

HOG-SVM: [0.875 0.975 0.925 0.925 0.975]
Mean = 93.50% Std = 3.74%

RAW-KNN: [0.85  0.8   0.9   0.825 0.85 ]
Mean = 84.50% Std = 3.32%

RAW-SVM: [0.9   0.825 0.925 0.9   0.875]
Mean = 88.50% Std = 3.39%

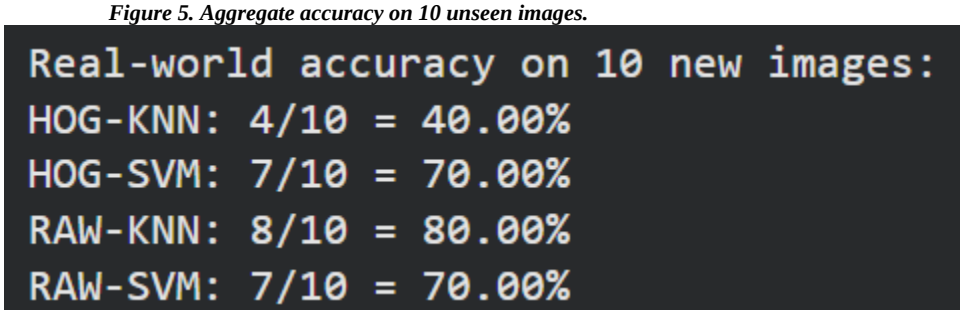
Highest mean accuracy: HOG-SVM (93.50%)
Most consistent: RAW-KNN (std 3.32%)
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Figure 4. Test accuracy and five-fold cross-validation comparison for all four models.

8. External-Image Evaluation

The four models were evaluated on a separate set of 10 unseen images. For privacy, this public portfolio version reports aggregate accuracy only. Individual test images, filenames and per-image predictions are intentionally omitted. The aggregate results are shown on the next page. Because the set contains only 10 images, it should be treated as a small generalisation check rather than a reliable estimate of real-world performance.

Figure 5. Aggregate accuracy on 10 unseen images.



Individual external test images are omitted from this public portfolio version.

9. Reflection

I learned that model accuracy depends on both the algorithm and the type of image features used.

My original idea that contrasting colours might help was partly correct. Raw colour pixels improved KNN, and Raw-KNN achieved the best real-world accuracy. HOG was more effective for SVM, which showed that colour alone was not enough.

One challenge was choosing the K value without guessing. I solved this by using 10-fold cross-validation. The results also showed that K = 1 may overfit, because HOG-KNN predicted all five new synthetic images as real.

The model could be improved by adding more varied training images representing a broader range of real and synthetic portrait styles.

Face detection and alignment could also be used to crop out backgrounds, clothing and accessories. This would help the model focus more on facial details.

Data augmentation could make the model more robust. Useful changes include adjusting brightness and contrast, converting images to grayscale, adding blur and resizing images.

I would also combine HOG features with colour histograms and tune the SVM and KNN settings through cross-validation.

A larger unseen test set with different cameras, lighting conditions and synthetic image generators would provide a more reliable measure of model generalisation.

10. Conclusion

This project successfully built and compared KNN and SVM models for classifying real and synthetic face images. The dataset was balanced, the images were preprocessed consistently, and both HOG and raw colour features were evaluated. HOG-SVM was the strongest model on the original dataset, reaching 92.50% test accuracy and 93.50% mean cross-validation accuracy. Raw-KNN performed best on the 10 new images with 80.00% accuracy. The project shows that evaluation should include both cross-validation and unseen real-world testing. Future improvements should focus on a larger dataset, more image variation and further model tuning.

11. Declaration of AI Use

AI tools were used to help explain code and organise the report structure. ChatGPT suggestion that was not followed was to add extra model pipelines and additional functions beyond the required project scope. This was not followed because the final code needed to remain simple and closely match the lessons taught in class.